Assessing the Safety of Cycling Networks: A Graph Theory Perspective

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***Abstract*— Current transport networks are by nature multi-modal and divided into categories (or layers) according to the prevailing means of transport. Among them, cycling networks are those where non-motorized vehicles such as bicycles, electric scooters, etc. are allowed. The key requirement for the use of cycling networks is that they must instill a feeling of safety, especially when space is shared with motorized vehicles. In this work, we use metrics from graph theory to quantify the connectivity of transport networks and apply them as a safety indicator for cycling networks. Specifically, we look into the following safety features: safety at intersections, safety along paths, connected components in the graph, and the presence of the same road segment on multiple paths. The proposed metrics are analyzed to understand which ones can be used as a basis for benchmarking the safety of any cycling transport network, and which ones are the most sensitive to changes in safety changes at the road-segment level. This paper analyzes the cycling transport network of the city of Grenoble, which is considered one of the densest cycling networks in France.**

***Index Terms*— Cycling transport network, graph theory, network connectivity metrics, centrality, safety.**

I. INTRODUCTION

Transport networks are the backbone of the mobility of people, products, and services from one location to another, which has greatly contributed to urbanization and globalization. However, globalization has brought consequences such as large amounts of carbon dioxide emissions, air pollution, the greenhouse effect, and more. This is why every day, people are invited to opt for the use of electric vehicles or soft means of transport, such as bicycles and scooters, for medium distances ($3\ \mathrm{km} > \mathrm{distance} > 1.5\ \mathrm{km}$), and walking or using skates for short distances (≤ 1.5 km) [1]. Bicycles dominate the soft transport networks because they take care of the environment while offering economic and social benefits [2], [3], [4] as they are considered healthy and comfortable.

Still, concerns about safety remain and may potentially slow down the uptake of soft transport means [5], [6], [7], [8]. Cities often limit themselves to showing map representations of the cycling network on their territory and to providing safety information at the level of each single road segment, thus giving very local information. Safety and perceived safety is not only local characteristic and much can be understood by looking at the interactions between modes and the network structure. Some studies have analyzed the interaction of motorized vehicles and bicycles, in which they share routes in the network [9], and it has been shown that in these scenarios, reducing the speed of cars does not guarantee an increase in safety for cyclists, since most bicycle accidents with cars are due to the lack of adequate separation between their respective lanes [10]. However, an infrastructure where bicycle and car lanes are separated from each other does increase the safety [11], [12], and lanes separated by physical barriers have been shown to completely reduce accidents caused by close passing cyclists, problems that still occur when bike lanes are only painted but not separated from motorized traffic. [13].

Furthermore, research efforts have been made to leverage graph theory to study different aspects of transport networks. For instance, there are metrics that have been used to measure the efficiency and level of service [14]. Other metrics have been developed to measure the resilience of a transport network topology when facing environmental risks [15]. In particular the betweenness centrality (section III-A) has been proved to be a suitable metric for computing the proportion of valid routes in a network after the failure of a road and for identifying key nodes in a network [16], [17]. Graph-theory connectivity indicators such as degree of connectivity and degree of cyclability, along with collision prediction models, have been used to model the safety of a bus system using census data [18], [19]. Finally, metrics have proved to be suitable indicators to estimate the importance of roads based on the number of cyclists riding on them. Nevertheless, there is no golden standard to analyze all the features and safety simultaneously in a network [20].

The first objective of this paper is to use graph theory metrics to assess the safety of soft transport networks in a more global and structural way, thus overcoming the limitations of the current road-segment level safety indicators. To this aim, we analyze the connectivity of soft transport networks based on pure network topology taking into account the safety information provided for each of their road segments (e.g. bicycle roads in good physical condition and separated from vehicles offer a high level of safety), in order to obtain a more global safety indicator. Specifically, we propose metrics to identify: the safety of an intersection, the average insecurity of traveling from one node to the rest of the graph, and the dispersion or integration of safe roads. We apply the suggested metrics to the cycling transport network of the city of Grenoble, which is considered one of the densest cycling networks in France, according to public data [1]. Finally, we employ the metrics to analyze the safety effects of a road

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[1]https://data.mobilites-m.fr/donnees

upgrading process, i.e., measure the difference in the increase in safety given by the global metrics when a road type is upgraded to one that offers greater safety to users. We then analyze which metrics are most sensitive to these upgrades in a step-wise manner. This can be helpful for governments and institutions as a first starting point on which type of road to focus on or to redesign the network topology in order to obtain a greater increase in overall safety.

The work is organized as follows. In Section II the local and global metrics are described for building our assessment framework. Section III introduces a descriptive metric that is used for the sensitive upgrading process analysis in Section IV where we also present our metrics evaluation results for analyzing the safety of the Grenoble cycling transport network. Finally, in Section V, we give some concluding remarks and discuss future work.

II. Local and Global Safety Metrics

In this section, we introduce global and local metrics to assess the safety of the cycling transport network. Let the considered network be described by a weighted graph $\mathcal{G} = (V, E, \omega)$, where V is a set of nodes (vertices), E is the set of links (edges), such that the nodes correspond to intersections or roads endpoints, links correspond to the roads, and $\omega_{u,v} \in \mathbb{Z}^+$ are the link weights assigned to the road (link) between the nodes u and v according to its road type. In this context, we assume that the type of road offers some safety to users, then we call $\omega_{u,v}$ as **bicycle road safety**, the higher the value, the safer the road. Typically, in existing European networks, we can find between 4 and 8 types of roads dedicated to cyclists. For instance, in the Grenoble network treated at the end of the paper, we can find 4 types, where roads are classified from 1 to 4, with 1 being the roads offering the lowest level of safety, and 4 the highest one. Throughout this paper, we will refer to the **safety level** of a road as the safety that its road type offers to the road user. Finally, we assume all roads are bidirectional, i.e. $\omega_{u,v} = \omega_{v,u}$, and hence the associated graph $\mathcal{G}$ is undirected.

A. Network intersection safety (Strength of nodes)

In graph theory for unweighted graphs, a way to measure the importance of a node is to use the "node-degree" [21] notion that measures the number of connections of a node (a local metric). The sum of all degrees of the nodes gives an aggregated measure of the degree of the associated graph (graph degree). For weighted graphs, this notion is named "node-strength" [22], and respectively the "graph-strength". In our framework, since weights indicate the level of safety that links offer to users, we can use this metric to identify the local safety level of an intersection (node), and the global nodal safety of the whole network. These metrics are defined in the following.

Given $v \in V$, its **local strength** is defined as:

$$s(v) = \sum_{e \in A_v} w_e, \tag{1}$$

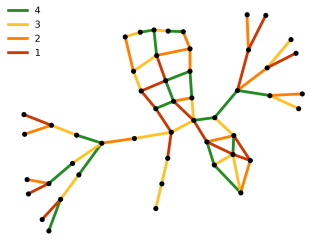


Fig. 1: Example network constructed to illustrate the proposed metrics assessment. Edges are colored according to their safety level.

with $A_v \subset E$ the set of adjacent edges to v. And through this strength, we obtain information about how safe an intersection in the network is. The **normalized local strength** is defined as:

$$s_N(v) = \frac{s(v)}{s_{FS}(v)} \in [0, 1]\,, \tag{2}$$

where s_{FS} is called the **local fully-safe network strength**, which is defined as the local strength considering the same set of nodes and edges as $\mathcal{G}$, but where all the edges are the type with the highest weight.

The **global strength** is defined as:

$$s_G = \frac{\sum_{v \in V} s(v)}{\sum_{v \in V} s_{FS}(v)} \in [0, 1]\,, \tag{3}$$

In Fig. 1 we have a network example where the links have a color assigned to their safety level that was determined randomly, and we compute the metrics in this subsection on it to show their utility, in particular for the normalized local strength metric see Fig. 2 (a) [2].

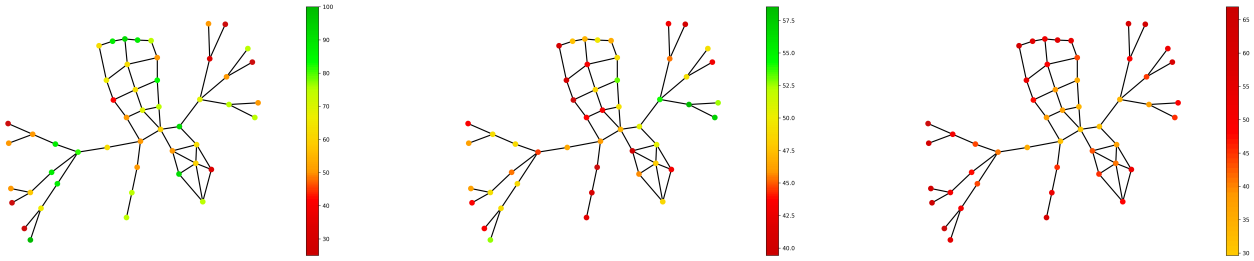
B. Network path safety (closeness centrality)

The problem of finding the shortest paths in terms of the number of roads between two points in a network is well known, as well as searching the paths that offer the smallest or largest sum of weights when working with a weighted graph. The closeness centrality metric is an average of how close a node is to the rest of the nodes in the graph in terms of shortest paths. However, in the case of the cycling transport network, it is known that users do not opt for the fastest paths, but for the routes with better facilities for bicycles, i.e., the ones that offer the highest safety, and are willing to travel at least 10% more in order to use the roads and sections that make them feel comfortable and safe [23]. Therefore, these metrics aim to find the average safety of going from one node in the network to the rest of the nodes.

Let $v \in V$, the **local closeness centrality** [24] of v in G is defined as follows:

$$c(v) = \frac{|V| - 1}{\sum_{u \in V} l_{u,v}}, \tag{4}$$

[2] All metrics in this work are normalized between 0 and 1, but for plotting and analysis purposes (section IV) they are multiplied by 100 to obtain percentage values

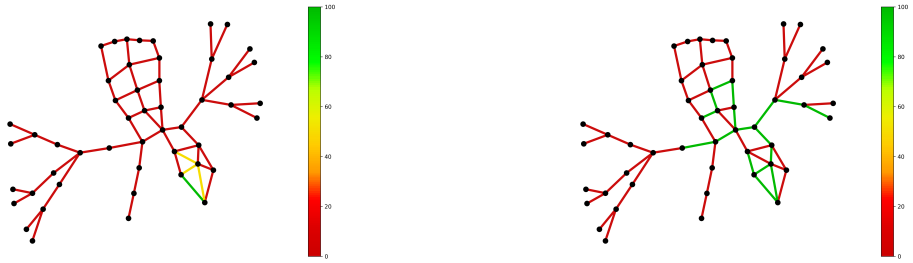


(a) Normalized local strength centrality, $s_G \approx 63.8\%$.

(b) Normalized local closeness centrality, $c_G \approx 46.6\%$.

(c) Normalized local farness centrality, $f_G \approx 49.3$ %.

Fig. 2: Illustration of the node-related metrics on the example graph. (a)-Nodes with higher values have more safe adjacent edges. (b)-Nodes with higher values indicate higher closeness to the rest of the nodes or are isolated by relatively safe and unavoidable paths. (c)-Nodes with higher values are the least centralized because they are the farthest away from the rest of the nodes.



(a) Normalized local edge clustering coefficient, $\varepsilon_G \approx 24.1$ %.

(b) Largest connected component and diameter, $d_G \approx 37.7$ %.

Fig. 3: Illustration of the edge-related metrics on the example graph. For both metrics, we define $\omega_0 = 3$. (a)-Edges with smaller values indicate that their adjacent edges are not integrated or are not sufficiently safe (weight $< \omega_0$), and therefore they disperse safe roads between unsafe ones. (b)-Edges with value 100 indicate that they belong to the largest connected component where edges have weight greater or equal to $\omega_0 = 3$, and edges with value 0 otherwise.

where $l_{u,v}$ is the sum of safety weights on the links conforming to the safest path between $u, v \in V$, with the safest path between u and v defined as the shortest path where the sum of the roads weights is the highest possible, $l_{u,u} = 0$ for all $u \in V$, and $|V|$ indicates the cardinality of V. Note that if $\mathcal{G}$ is disconnected and u and v are in different connected components, then $l_{u,v}$ is equal to 0. The **normalized local closeness centrality** is defined as:

$$c_N(v) = \frac{c(v)}{c_{FC}(v)} \in [0, 1]\,, \tag{5}$$

where $c_{FG}(v)$ is the **local fully-safe network closeness centrality** of node v defined as the closeness centrality of the node v in the network where all the edges have the highest safety level. And then the **global safety closeness** (c_G) is defined as:

$$c_G = \frac{\sum_{v \in V} c(v)}{\sum_{v \in V} c_{FC}(v)} \in [0, 1]\,. \tag{6}$$

In Fig. 2 (b), nodes are colored according to their normalized local closeness centrality, that is, the percentage of safety to travel from intersections or the endpoint of roads to any other in the graph compared to the same analysis in the graph where all the edges have the highest safety level.

C. Network path insecurity (farness centrality)

Sometimes, we may not wish to look for measures of closeness in a network, but rather the opposite, measures of "farness". In a classical way, the farness centrality is defined as the inverse of closeness centrality [25], and it gives an average of how far it is to go from one node in the network to any other. In terms of safety, this concept translates as the average insecurity of using a bicycle from one point to any other in the network.

The farness centrality is not defined for not connected graphs, therefore for our purposes, it is first necessary to identify the connected subgraphs, which are here named $\mathcal{H}_c \subset \mathcal{G}$ for $c \in \{1, \ldots, n_c\}$ where n_c is the number of connected subgraphs in $\mathcal{G}$. We can now define the farness centrality as follows.

We denote $V_{\mathcal{H}_c}$ as the set of nodes in $\mathcal{H}_c$, and $|V_{\mathcal{H}_c}|$ as the cardinality of $V_{\mathcal{H}_c}$, then, $\forall c \in \{1, \ldots, n_c\}$ we define the **local farness centrality** of a node $v \in V_{\mathcal{H}_c}$ as :

$$f(v) = \frac{1}{|V_{\mathcal{H}_c}| - 1} \sum_{u \in V_{\mathcal{H}_c}} l_{u,v}. \tag{7}$$

The **normalized local farness** is defined as:

$$f_N(v) = \frac{f(v)}{\text{diameter}(\mathcal{H}_c^*)} \in [0,1] \tag{8}$$

In this case, we normalize not with respect to the safest-case scenario, but with respect to an unsafest-case scenario where all edges have the lowest weights corresponding to the least safe roads. That is, we suppose that in the least safe scenario, all paths between two different nodes in the graph have a distance equal to the diameter of $\mathcal{H}_c^*$, where $\mathcal{H}_c^*$ is the connected subgraph $\mathcal{H}_c$ where all edges have the lowest weight.

And then the **global farness centrality** of $\mathcal{H}_c$ is defined as:

$$f_G = \frac{1}{n_c} \sum_{c \in \{1,\dots,n_c\}} f_{\mathcal{H}_c} \in [0,1]\,, \tag{9}$$

where

$$f_{\mathcal{H}_c} = \frac{\sum_{v \in V_{\mathcal{H}_c}} f(v)}{|V_{\mathcal{H}_c}| \cdot \text{diameter}(\mathcal{H}_c^*)} \in [0,1]\,. \tag{10}$$

where $|V_{\mathcal{H}_c}|$ is the cardinality of the set $V_{\mathcal{H}_c}$. In Fig. 2 (c) the nodes are colored with their normalized local farness, that is the percentage of insecurity of traveling from that node to any other reachable node in the current graph compared to the same analysis in the network where all paths are of the least safe type.

D. Network distance to path safety (edge clustering coefficient)

In general, knowing if edges with similar weights are connected to each other or if they tend to connect with edges with dissimilar weights gives us valuable information about a weighted graph. The assortative coefficient [26] measures this feature in a normalized range $[0,1]$, with 0 describing the least assorted, while 1 gives the maximum value in Eq. (11).

The **assortative coefficient**, α is defined as:

$$\alpha_G = \frac{1}{2}\left(\frac{a_1 - a_2^2}{a_3 - a_2^2} + 1\right) \in [0,1] \tag{11}$$

with,

$$a_1 = \frac{1}{S} \sum_{(i,j)\in E} (\omega_{i,j} \prod_{v \in \{i,j\}} s(v)) \tag{12}$$

$$a_2 = \frac{1}{2S} \sum_{(i,j)\in E} (\omega_{i,j} \sum_{v \in \{i,j\}} s(v)) \tag{13}$$

$$a_3 = \frac{1}{2S} \sum_{(i,j)\in E} (\omega_{i,j} \sum_{v \in \{i,j\}} s(v)^2) \tag{14}$$

where $S = \sum_{v \in V} s(v)$. This metric provides helpful knowledge about whether safe roads tend to connect with other safe roads or tend to connect with unsafe roads and vice-versa. An assortative coefficient smaller than $\frac{1}{2}$ means that safe roads tend to be dispersed and disconnected between unsafe roads, while a coefficient greater than $\frac{1}{2}$ says that the network tends to have safe roads together and unsafe roads apart. However, the assortative coefficient is a completely global measure of how edges with similar weights tend to connect, additionally, sometimes it is not clear how to define the similarity level between the weight of the edges. Fortunately, there are other local metrics, such as edge clustering coefficient [27], [28], that allow us to analyze similar phenomena, but with more focus on how an edge integrates or disperses adjacent roads. Also, edge clustering allows us to define when the weights of the edges are similar or dissimilar enough, see Eq. (15) and Fig. 4.

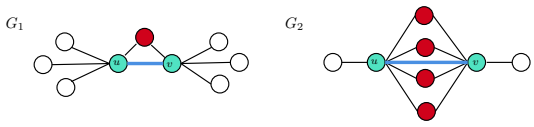


Fig. 4: The common neighbors of u and v are shown in red. The edge (u,v) in G_1 has a lower $\varepsilon(u,v)$ than in G_2. Hence, we can see that (u,v) in G_1 disperses adjacent edges whereas in G_2 it integrates them.

Given $v \in V$, we define N_v as the set of neighbors of v, N_u as the set of neighbors of u, and $\forall v \in V$ we call $N_v^{\omega_0} := \{e(v,u) \in E | w_{v,u} \geq \omega_0, u \in N_v\}$ as the set of the neighbors of v sharing the roads with safety level higher or equal than $\omega_0 \in \mathbb{N} \cup \{0\}$ and $\omega_0 < \max \omega_{v_1,v_2}$ for all $v_1, v_2 \in V$. Now, we define the **local edge clustering coefficient** of edge $e = (u,v) \in E$ as follows:

$$\varepsilon(u,v) = \frac{|N_u^{\omega_0} \cap N_v^{\omega_0}|}{|N_u \cup N_v| - 2}, \tag{15}$$

And $\varepsilon(u,v) = 0$ if $|N_u \cup N_v| - 2 = 0$. The **normalized local edge clustering coefficient** is defined as:

$$\varepsilon_N(u,v) = \frac{\varepsilon(u,v)}{\varepsilon_{FE}(u,v)} \in [0,1]\,, \tag{16}$$

where $\varepsilon_{FE}(u,v)$ is the **local fully-safe network edge clustering coefficient**, which is defined as the edge clustering coefficient of the edge (u,v) in the same graph G, but where all the edges have the highest weight, and then offer the highest safety level. If $\varepsilon_{FE}(u,v) = 0$, then $\varepsilon_N(u,v)$ is defined as 0. The **global edge clustering coefficient** of $\mathcal{G}$ is defined for all $e \in E$, as:

$$\varepsilon_G = \frac{\sum_e \varepsilon(e)}{\sum_e \varepsilon_N(e)} \in [0,1]\,. \tag{17}$$

In Fig. 3 (a) edges are colored according to their normalized local edge clustering coefficient. This means that the image shows the ratio of how edges cluster their safest adjacent edges compared to the same analysis in the network where all the edges have the highest safety level.

E. Network safest connected roads (Largest connected component and diameter)

People will prefer to travel through a network with connected safest paths instead of a network with a high number of safest roads but dispersed among risky roads. A metric reflecting the size of the safest path connectivity in the network can be defined as the ratio between 1) the number of nodes in the safest connected component of the network plus its diameter, and 2) the number of nodes in the safest

connected component and the diameter of the graph where the edges have the highest safety level.

Largest connected component and diameter Let C_{ω_0} the set of edges with weight greater or equal to ω_0, with ω_0 as in section II-D. And let be $\mathcal{H}$ the largest connected graph induce by C_{ω_0}, let $|V_{\mathcal{H}}|$ be the number of nodes of $\mathcal{H}$. And $\mathcal{H}_{FS}$ is defined as $\mathcal{H}$ but in the case where all the edges have the highest weight i.e., the network where all the edges are the safest ones. Then we define:

$$d_G = \frac{|V_{\mathcal{H}}| + \text{diameter}(\mathcal{H})}{|V_{\mathcal{H}_{FS}}| + \text{diameter}(\mathcal{H}_{FS})} \in [0, 1]\,, \tag{18}$$

This metric is a global measure of the proportion of vertices in the safer connected components and how large it is through the network compared with the case where every road in the largest connected component of the network would have the highest weight and then offer the highest safety level.

In Fig. 3 (b), we identify the safest and largest connected component of the current network.

III. Topology-dependent Local Safety Metrics

In graph theory, there are a large number of metrics describing different features that relate to the nodes and links characteristics as the ones discussed in the previous section. The metric that will be discussed in this section describes the topological structure of the network as a whole. In our context of studying safety in cycling networks, analyzing the topological characteristics of the network together with their related link weights allows us to assess the relative importance in terms of the safety of a road with respect to all possible paths in the network traversing this road.

A. Network road safety (Betweenness Centrality)

Betweenness centrality is a measure that characterizes the number of times an edge appears in the shortest paths between any two points in a graph. It describes how important an edge is in the whole graph in terms of the shortest paths that pass through it [15]. In our context this metric measures the frequency with which a road is part of the safest paths in the whole network.

Let $\sigma_{u,v}$, be the number of safest paths from u and v, for all $u, v \in V$. Let $\sigma_{u,v}(e)$ be the number of these safest paths passing through edge e. We define the **betweenness centrality** of edge e in graph G as:

$$b(e) \colon = \sum_{u,v \in V} \frac{\sigma_{uv}(e)}{\sigma_{uv}}, \tag{19}$$

The **normalized betweenness centrality** is defined as:

$$b_N(e) = \frac{b(e)}{\sum_{e \in E} b(e)} \in [0, 1]\,, \tag{20}$$

Note that except for very special cases of networks with linear (and trivial) topologies, the normalized metrics will never reach their maximum, but it is still useful to weigh the relative importance of a link in a network. In the example shown in Fig. 5, edges are colored according to the value of $b_N(e)$. Note that edges with higher values are not necessarily the safest but several safe paths cross them. In that, edges with higher values become the most important ones in relative importance to the others.

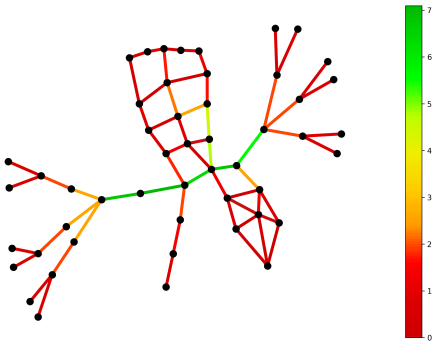


Fig. 5: Illustration of the normalized betweenness centrality on the example graph. The higher the value of an edge is, the more frequently the edge appears in the safest paths between two different nodes.

IV. Safety assessing of the Grenoble Bikes Network

In this section, we evaluate the previously introduced metrics on the Grenoble metropolitan cycling network. We first present the physical bike network layout which is already split into several sub-networks according to different types of infrastructure and degree of separation from the motorized transport network. Then we transform this network into a weighted graph and compute the global and local metrics associated with it. Finally, we analyze different scenarios in order to assess the sensitivity of the proposed metrics to different plans for upgrading the network and improving road safety.

A. The Grenoble cycling network

The Grenoble soft transport network is one of the densest cycling networks in France[3]. The roads are divided into four different types or categories according to their characteristics and degree of separation from other transport modes, as detailed in Table I, thus giving four different sub-networks. Their layout is shown in Fig. 6.

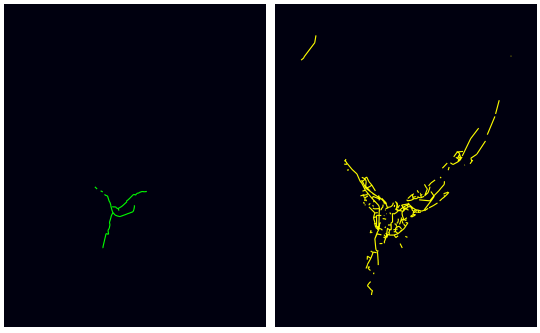
B. Network upgrade scenario

The mathematical graph, $\mathcal{G} = (V, E, \omega)$, being $\mathcal{G} = \mathcal{G}_1 + \mathcal{G}_2 + \mathcal{G}_3 + \mathcal{G}_4$, (+ means union of the sets of nodes and edges), associated with the soft transport of Grenoble, has been created using Python libraries[4], that transform the public GeoJSON data from the Grenoble cycling network[5], into a mathematical object. This gives a graph with the following characteristics: $|V| = 1968$ and $|E| = 2227$, and provides the adjacency matrix $\mathbf{A} \in \mathbb{R}^{1968 \times 1968}$ of $\mathcal{G}$.

[3]https://barometre.parlons-velo.fr/2021/
[4]https://github.com/leunamCampero/IEEEITSC-2023
[5]https://data.mobilites-m.fr/donnees

Type Road	Safety level	Description
Chronovelo ($\mathcal{G}_4$)	High = 4	Exclusive and high-quality free-ways for cyclists separated from the automobile transport network, offering elevated safety to users.
Veloseparatif ($\mathcal{G}_3$)	Medium High =3	Highways of acceptable quality and usually separated from mixing with the automobile network, but not always.
Veloconseille ($\mathcal{G}_2$)	Medium Low =2	Deteriorated and narrow bicycle roads that are not necessarily separated from the motorized vehicles.
Velodifficile ($\mathcal{G}_1$)	Low=1	Deteriorated and narrow bicycle roads mixed with vehicles network.

TABLE I: Soft transport roads description per sub-network.



(a) $\mathcal{G}_4$ (b) $\mathcal{G}_3$

(c) $\mathcal{G}_2$ (d) $\mathcal{G}_1$

Fig. 6: Graph representation of the different sub-graphs in Grenoble cycling network, the number of edges (roads) in $\mathcal{G}_4, \mathcal{G}_3, \mathcal{G}_2$ and $\mathcal{G}_1$ is $108, 1063, 769$ and 287 respectively.

This will allow us to compute the metrics defined before. In particular, we will be using the set of global metrics defined by the vector, m,

$$m = [s_G, c_G, f_G, \varepsilon_G, d_G], \tag{21}$$

for the edge clustering coefficient and largest connected component analysis on the Grenoble soft transport network, we define $\omega_0 = 3$, i.e. we suppose that the safe-enough types of roads are the roads in $\mathcal{G}_3$ and $\mathcal{G}_4$.

In order to evaluate the significance of these metrics, we propose to assess the impact of a safety upgrade program that public bodies might wish to implement in the city of Grenoble as a result of public investment. Therefore, decision-makers may find themselves in the difficult position of deciding which links of the cycling network to upgrade first in order to maximize the positive effects on the overall bicycle uptake and safety perception. In the following, we consider, two scenarios to asses such an upgrading process.

1) Network multi-level upgrading: Consider a first scenario where we will be "upgrading" only one sub-network at a time and measure the obtained safety improvement via a Euclidean norm, so as to identify what upgrade step is the most significant. We use the notation m_i to indicate the value of m through the upgrading process, which is shown in Table II:

Upg / Lvl	$i=0$	$i=1$	$i=2$	$i=3$
$\mathcal{G}_4$	4	4	4	4
$\mathcal{G}_3$	3	3	3	4
$\mathcal{G}_2$	2	2	3	2
$\mathcal{G}_1$	1	2	1	1

TABLE II: Values in the table show the upgrade safety levels of the multi-level upgrading scenario. The original network is indicated with $i = 0$.

Δm_1	Δm_2	Δm_3
3.5 %	86.0 %	13.0 %

TABLE III: Euclidean norm of the vectors obtained through the multi-level upgrading scenario in Table II.

To measure incremental improvements as the upgrading process occurs, we introduce the upgrade improvement measure as the norm of the difference between successive multi-level upgrades, i.e. $\Delta m_i = ||m_0 - m_i||$, $i = 1, 2, 3$. Table III shows the obtained improvement. As we can observe from this table, the largest payoff is obtained in Δm_2, meaning that the highest safety improvement is obtained when upgrading the roads in $\mathcal{G}_2$. This suggests that, if an upgrade process of this type is envisioned, investments should be concentrated to update all the roads of subgraph $\mathcal{G}_2$.

2) Network single-level upgrading: In section IV-B we find that subgraph $\mathcal{G}_2$ provides the highest safety payoff for the upgrading process. This outcome is expected as $\mathcal{G}_2$ has the largest number of links. However, it is important to consider that the government or institutions cannot change all road types overnight. Therefore, we need to determine a gradual approach to upgrade the roads in $\mathcal{G}_2$, eventually reaching the safety payoff when they are all the same type as $\mathcal{G}_3$.

To achieve this, we propose sorting the edges in $\mathcal{G}_2$ in descending order [6] based on their betweenness centrality, which indicates their frequency in the total number of safest paths in the network. Previous studies have shown that betweenness centrality is a suitable metric for identifying important edges and key nodes [16], [17]. Once the edges are sorted, we conduct safety analyses using the metrics while gradually increasing the percentage of upgraded roads (starting with 5%, then 10%, 15%, up to 100%). We repeat

[6]**Descending order** means edges are arranged from those that appear most frequently on all the safest paths to those that appear less frequently, meanwhile **ascending order** means the order in the opposite manner

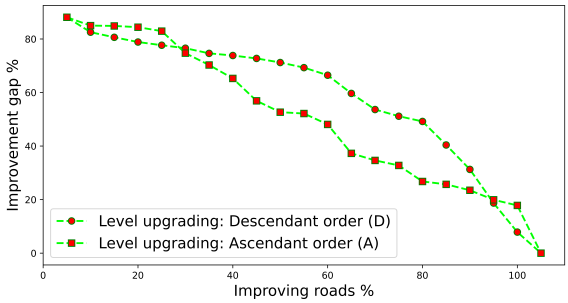


Fig. 7: Global impact of gradually upgrading $\mathcal{G}_2$ from level 2 to level 3. The improvement gap is the difference in percent between the value of the metrics during the gradual upgrade and their value when all the roads are upgraded from level 2 to level 3.

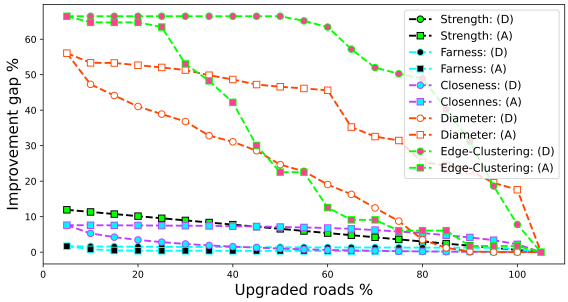


Fig. 8: Individual metrics variation when gradually upgrading $\mathcal{G}_2$ from level 2 to level 3. The improvement gap is the difference in percent between the value of the metrics during the gradual upgrade and their value when all the roads are upgraded from level 2 to level 3.

this analysis, but this time with the roads sorted in ascending order (see Fig. 7). Interestingly, in the initial stages of the gradual upgrading, it appears more beneficial to upgrade the roads from the highest to the lowest frequency on the safest paths based on betweenness centrality. However, in most of the later upgrade steps, it is more advantageous to upgrade the roads from the lowest to the highest frequency on the safest paths.

Overall, our findings suggest a systematic and efficient approach to upgrading the roads in $\mathcal{G}_2$ towards achieving the desired safety improvements.

In Figure 8, we show the impact of a gradual upgrade on all the metrics, individually. Depending on the desired safety enhancements, we need to follow a particular metric and ordering approach. For example, if our goal is to rapidly improve the overall safety proximity (i.e. closeness centrality) when upgrading all roads in $\mathcal{G}_2$ to the level of $\mathcal{G}_3$, it is recommended to prioritize the roads based on their betweenness centrality, starting from the highest to the lowest frequency on the safest paths. On the other hand, if the objective is to integrate the safe roads of the network and separate them from the unsafe ones (i.e. edge clustering), an ascending order of betweenness centrality is necessary. It is worth noting that the strength value remains the same for both ordering approaches, as defined.

Based on the analysis of the two scenarios, it can be inferred that the upgrading process should primarily target the sub-network $\mathcal{G}_2$. A significant improvement can be achieved already by focusing on approximately 60% of the roads of $\mathcal{G}_2$, identified by sorting in ascending order according to the betweenness centrality. It is important to note that the proposed methodology enables the examination of alternative upgrading combinations, opening up the possibility to explore numerous other scenarios and options for further study. In other words, this work lays the foundations for future work involving optimization. Some of the proposed metrics (e.g. edge clustering) have shown a high sensitivity to changes in the edge weights, which is very promising if one wants to use such metrics as an objective function in a topology optimization problem.

V. Conclusion

This paper proposes the application of graph theory to assess the safety of soft transport networks in urban environments. We utilize graph theory metrics, that are responsive to changes in graph weights (i.e. safety value on each edge), to understand how road-level information is transformed into neighborhood- or global-level information whenever a change in the topology occurs. Therefore, changes in weights can be directly correlated with perceived safety improvements across the entire network. The influential metrics primarily examine the size and connectivity of roads that offer greater safety to individuals. Additionally, they assess whether safe roads are interconnected or scattered among unsafe roads.

Our metrics replicate and confirm the general assumption that cyclists and users of the soft transport network are less inclined to travel on it when there are isolated safe roads. Conversely, they are more likely to utilize soft transport if safe roads are well-connected and distributed throughout the network, even if the number of safer roads is relatively smaller. Therefore, the proposed metrics could be used as a proxy for global cycling network safety.

In our analysis of the cycling transport network of the city of Grenoble, we discovered that the most significant safety improvement would result from upgrading all roads ranked as the second least safe to the next safety level. Moreover, we identified the possibility of achieving progressive safety enhancements by gradually upgrading roads within a network, enabling focused investments in prominent safety improvements.

In future work, we plan not only to integrate the proposed metrics in a topology optimization framework but also to compare our safety analysis with public survey data about perceived safety along paths in cycling networks.

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